

Big Data Exam Crib Sheet 2026

February 2026

1 Apache Spark Functions

- **map(func)**: Return a new distributed dataset formed by passing each element of the source through a function func.
- **filter(func)**: Return a new dataset formed by selecting those elements of the source on which func returns true.
- **flatMap(func)**: Similar to map, but each input item can be mapped to 0 or more output items (so func should return a Seq rather than a single item).
- **mapPartitions(func)**: Similar to map, but runs separately on each partition (block) of the RDD, so func must be of type $\text{Iterator}_i T_i \Rightarrow \text{Iterator}_i U_i$ when running on an RDD of type T.
- **mapPartitionsWithIndex(func)**: Similar to mapPartitions, but also provides func with an integer value representing the index of the partition, so func must be of type $(\text{Int}, \text{Iterator}_i T_i) \Rightarrow \text{Iterator}_i U_i$ when running on an RDD of type T.
- **union(otherDataset)**: Return a new dataset that contains the union of the elements in the source dataset and the argument.
- **intersection(otherDataset)**: Return a new RDD that contains the intersection of elements in the source dataset and the argument.
- **distinct([numPartitions])**: Converts an $\text{RDD}_i V_i$ to an $\text{RDD}_i K, V_i$ using a function func, which for a record generates a key, then groups the records by key into partitions.
- **groupByKey([numPartitions])**: When called on a dataset of (K, V) pairs, returns a dataset of $(K, \text{Iterable}_i V_i)$ pairs.
- **reduceByKey(func, [numPartitions])**: When called on a dataset of (K, V) pairs, returns a dataset of (K, V) pairs where the values for each key are aggregated using the given reduce function func, which must be of type $(V, V) \Rightarrow V$. Like in groupByKey, the number of reduce tasks is configurable through an optional second argument.

- **aggregateByKey(zeroValue)(seqOp, combOp, [numPartitions]):** When called on a dataset of (K, V) pairs, returns a dataset of (K, U) pairs where the values for each key are aggregated using the given combine functions and a neutral "zero" value. Allows an aggregated value type that is different than the input value type, while avoiding unnecessary allocations. Like in groupByKey, the number of reduce tasks is configurable through an optional second argument.
- **sortByKey([ascending], [numPartitions]):** When called on a dataset of (K, V) pairs where K implements Ordered, returns a dataset of (K, V) pairs sorted by keys in ascending or descending order, as specified in the boolean ascending argument.
- **join(otherDataset, [numPartitions]):** When called on datasets of type (K, V) and (K, W), returns a dataset of (K, (V, W)) pairs with all pairs of elements for each key. Outer joins are supported through leftOuterJoin, rightOuterJoin, and fullOuterJoin.
- **cogroup(otherDataset, [numPartitions]):** When called on datasets of type (K, V) and (K, W), returns a dataset of (K, (Iterable[V], Iterable[W])) tuples. This operation is also called groupWith.